Executive Summary

Decentralized Academic Record Management and Credential Verification System Using Blockchain and Web3 Technology

SP301 Special Project Work · Kasala Yashas Reddy (24STUCHH010288), Kurri Bhavesh Kumar (24STUCHH010966), Vegesna Sujit Varma (24STUCHH010891), Mantena Laxmi Kishore Varma (24STUCHH010060) · Guide: Dr. D. Krishna Madhuri · ICFAI Foundation for Higher Education, Hyderabad

Measurements reproduced on Tue, 13 Oct 2026 05:58:11 GMT (local Hardhat node (automining)).

1. What is happening (how the system works)

The system is a full-stack Student Information System (SIS) where **managing academic records is the product** and **trustless verification is one integrated feature**. Academic data lives in MySQL 8 (the mutable source of truth); the blockchain stores only immutable evidence.

- **Issuance.** An ISSUER/ADMIN wallet session compiles a canonical, key-sorted transcript JSON from live grades, computes a SHA-256 digest (of an uploaded PDF's exact bytes when supplied, otherwise of the canonical JSON), stores the record as DRAFT, anchors the digest + IPFS CID + issuer + timestamp on the AcademicRecordManager contract, and only then marks it ACTIVE. A failed transaction can never appear as issued (DRAFT + retry endpoint).
- **Lifecycle.** Records transition Active → Revoked or Active → Superseded. A correction (*supersession*) publishes the new record and retires the old one in a *single atomic transaction*, with the replacement link stored on-chain.
- **Verification (integrated feature).** Anyone verifies by record ID or by dropping the document: the digest is computed in the browser, looked up through a read-only `eth_call` (zero gas, no wallet) and cross-checked against MySQL. The response reports database status, chain status, an explicit consistency flag (status *and* digest must match — direct database tampering is flagged), and the issuer credential signature. Unknown IDs return 404; a chain outage is reported as *unavailable*, never as *tampered*.
- **Identity & RBAC.** Password-less MetaMask `personal_sign` login produces stateless 12-hour session tokens (no server secret). Roles ADMIN/ISSUER/STUDENT/VERIFIER are enforced by the API *and* independently by the contract's issuer whitelist — the ledger cannot be written by an unauthorized wallet even if the API were bypassed.
- **New in this build.** QR codes and shareable `/verify/<id>` links; browser-side (MetaMask) signing of issuance/revocation transactions with server-signer fallback; AES-256-GCM encryption at rest for stored PDFs; issuer signature over the record digest, re-verified publicly; a 23-mutation tamper suite and an automated chain-outage failure-path test.

2. What is used (stack and evidence)

Layer	Technology	Purpose
Frontend	React 18.3 + Vite 5, custom dark CSS, ethers v6, qrcode	Dashboard, Records, Students, student portal, public Verify (5 pages)
Backend	Node.js 26 + Express 4.21, mysql2, ethers v6	REST API, transcript compilation, hashing, sessions, chain coordination, audit trail

Database	MySQL 8.0 (6 tables)	users, programs, courses, enrollments, academic_records, audit_logs
Smart contract	Solidity 0.8.24, Hardhat 2.29	issuance, lifecycle, atomic supersession, RBAC whitelist, public view verification
Network	local Hardhat node (automining); Sepolia configured	immutable state + events
Storage	encrypted uploads (AES-256-GCM) + IPFS via Pinata (optional)	documents at rest; permanent CID when configured

Measured results (docs/measurements.json, Tue, 13 Oct 2026 05:58:11 GMT)

Operation	Trials	Min	Mean	Max	Gas
Public verification (GET /api/verify, eth_call + DB cross-check)	10	45 ms	54.7 ms	69 ms	0 (read-only)
Record issuance (compile → hash → MySQL → confirmed tx)	3	211 ms	304 ms	384 ms	1,01,548
Revocation (status update tx)	1	—	331 ms	—	53,179
Correction / supersession (one atomic tx)	1	—	174 ms	—	1,50,684
Issuance throughput (20-tx batch, pre-signed, raw submission)	20	194.2 tx/s (target > 20 — met)			1,01,560/tx

Verification evidence (all repeatable)

- 14/14 Hardhat unit tests (RBAC, issuance, duplicate-ID rejection, lifecycle, view verification)
- 52/52 end-to-end REST assertions (auth, 401/403, issuance, verification, tamper rejection, revocation, supersession, wallet-signed issuance + revocation, encrypted documents, signature checks, scoping)
- 10/10 chain-service assertions (issue → verify → supersede → revoke round-trip)
- 23/23 tamper-suite mutations rejected (bit flips, truncation, extension, header corruption, block zeroing, forged transcript, wrong file) + direct database tampering flagged by the DB/chain cross-check
- 7/7 chain-outage assertions (dead RPC → "unavailable", never "tampered"; 404 semantics preserved)
- Production client build succeeds; browser check: dashboard, QR deep-link verification and 404 path render with a clean console

Honest limitations. Sepolia deployment and Pinata pinning are fully wired (env-configured) but were not exercised here — no funded testnet key or Pinata JWT was available. The credential signature is an issuer wallet signature over the record digest, not a W3C Data Integrity proof; no standards compliance is claimed. QR codes link to the verification page; camera-based QR scanning is not implemented. Wall-clock timings are from a local automining node; public-network timings will differ.

3. What more can be built

- **W3C Verifiable Credentials 2.0 + DID methods** — issue credentials as full VCs with Data Integrity proofs and a status list for revocation, enabling ecosystem-wide interoperability (EBSI, 1EdTech Open Badges).

- **Public-network rollout** — Sepolia → mainnet/L2 anchoring (Polygon/Arbitrum for cost), multi-institution issuer governance (multisig admin, per-faculty whitelists), and measured public-network latencies.
- **QR scanning and offline verification** — camera-based QR capture, a PWA mobile wallet, and an offline-capable verifier that caches Merkle proofs.
- **Zero-knowledge selective disclosure** — prove "GPA ≥ 3.5" or "degree conferred" without revealing the full transcript (ZKBAR-V-style), keeping privacy while retaining verifiability.
- **IPFS hardening** — pin via Pinata/web3.storage with redundancy, encrypted CID publication, and gateway health monitoring.
- **Registrar tooling** — batch issuance CSV import, cohort statistics, LMS/ERP integration (SIS connectors), and automated degree-audit checks before issuance.
- **Security operations** — key-ceremony/HSM for the server signer, key rotation, alerting on anchor failures, and a public audit explorer over contract events.

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